

EE 226: Semiconductor Devices

Lecture 7: Finding Carrier Concentration – Part 2

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Course Coverage

1. What is current? $I = \frac{dQ}{dt}$ Module 1

1. $Q \Rightarrow$ electrons, holes $\Rightarrow$ How many are there?

2. How do they flow in a given device?

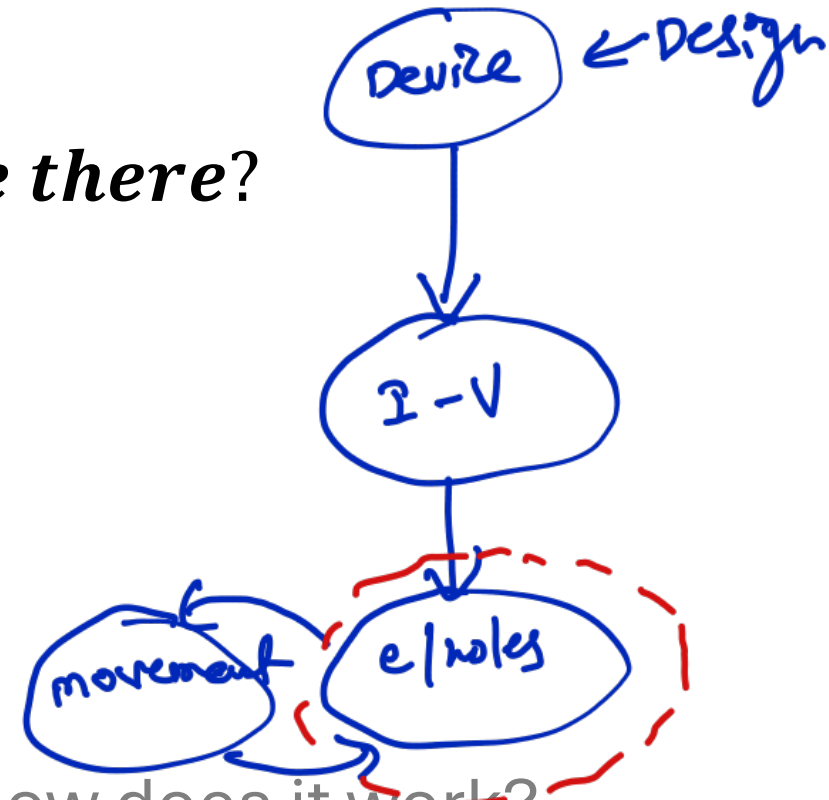
3. Any other dependencies

2. How does a diode work? Module 2

1. All the questions raised in previous slide.

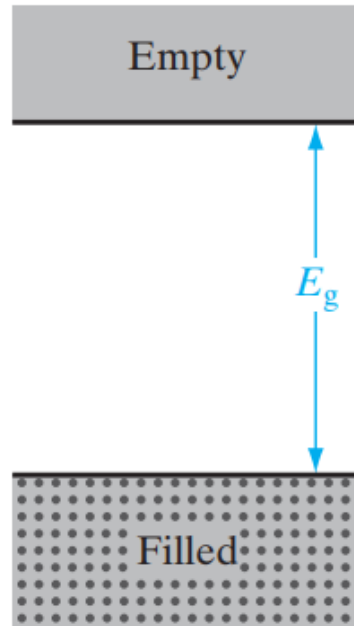
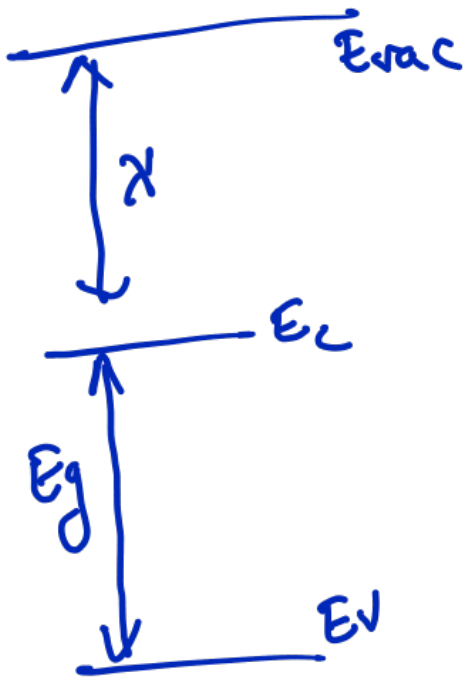
3. Why do we need a 3 (or 4)-terminal device & how does it work?
Module 3, Module 4

4. Need and operation of memory devices?
Module 5

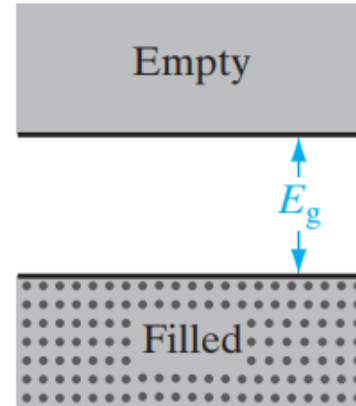


Recap: Typical Band Structures

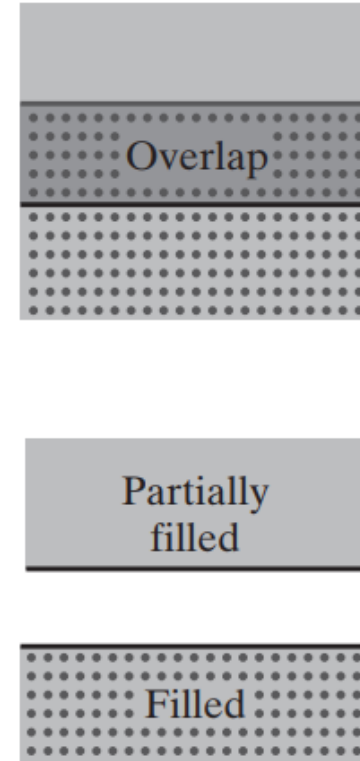
At 0K temperature



Insulator



Semiconductor



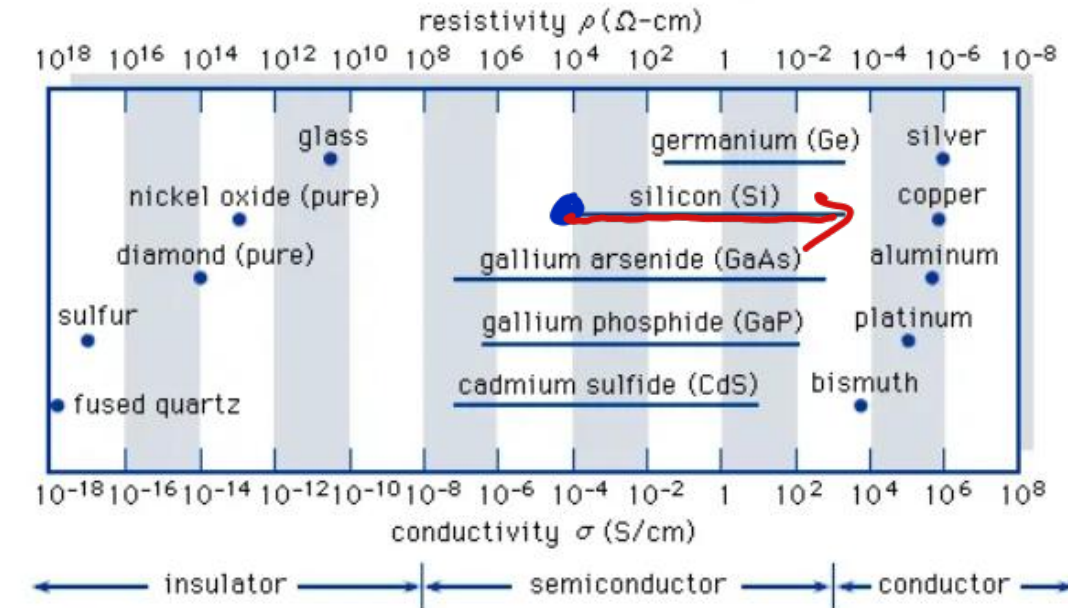
Metal



Are these intrinsic carriers enough for conduction?

	Si	Ge	GaAs
Energy Gap at 300K (eV)	1.12	0.66	1.424
Intrinsic Carrier Concentration (cm^{-3})	1.45×10^{10}	2.4×10^{13}	1.79×10^6

- The free carrier concentration of intrinsic semiconductor provides low conductivity (insulating) at room temperature
- Need some mechanism to increase the free carrier concentration and increase conductivity



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How to increase carrier concentration?

1. External energy Sources
 1. Temperature, light etc

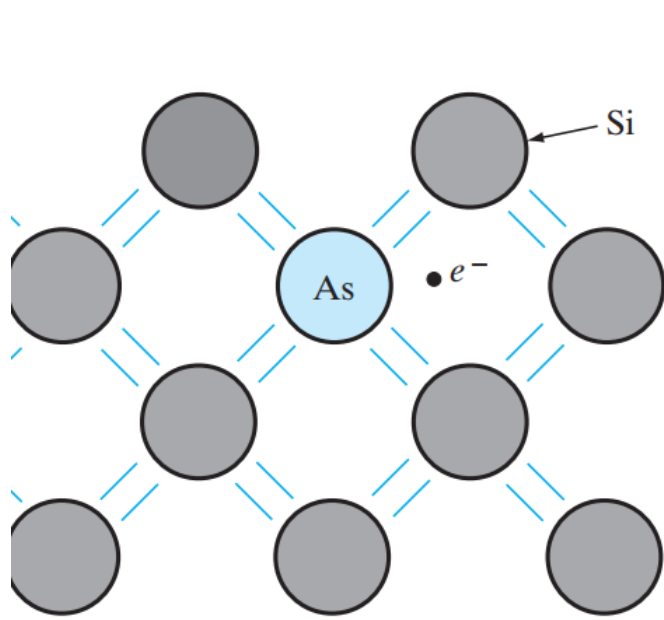
2. Doping – Adding Impurity atoms



$$\begin{array}{l} \text{300K} \\ \text{1 free } e^- / 10^{12} \text{ Si atoms} \sim 10^{-4} \text{ (S/cm)} \\ \text{1000K} \\ \text{1 free } e^- / 10^4 \text{ Si atoms} \sim 10^4 \text{ (S/cm)} \end{array}$$

A red arrow points from the first expression to the second, indicating a transition or comparison.

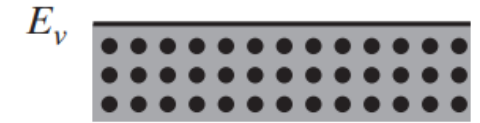
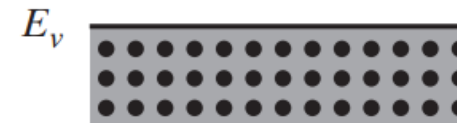
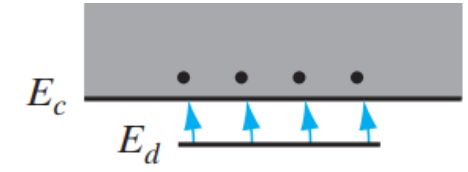
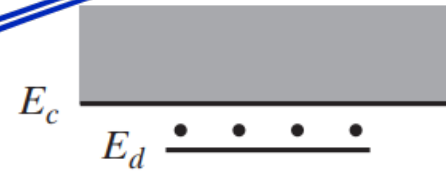
Extrinsic Semiconductor (n-type)



$\sim 10^{18} / \text{cm}^3$
 ~ 1 impurity atom / 10^4 Si atoms

Donors
Phosphorus (P)
Arsenic (As)
Antimony (Sb)

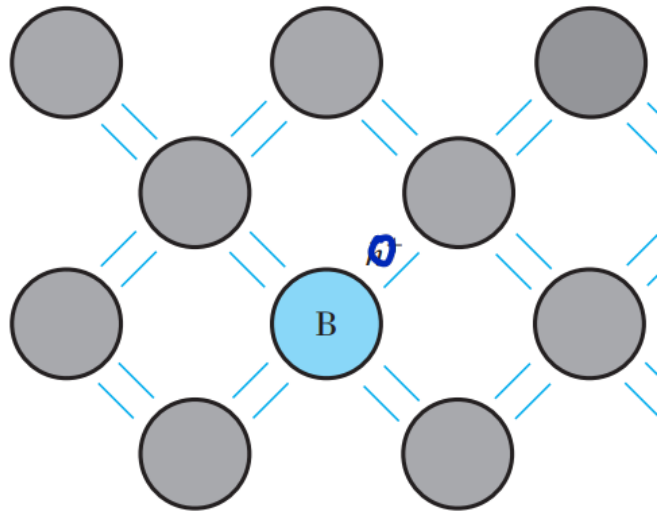
Impurity Atoms
Pentavalent



Majority carriers: Electrons
Minority carriers: Holes

How to calculate the intrinsic carrier concentration (n_i)?
How to find majority carrier concentration?
How to calculate minority carrier concentration?

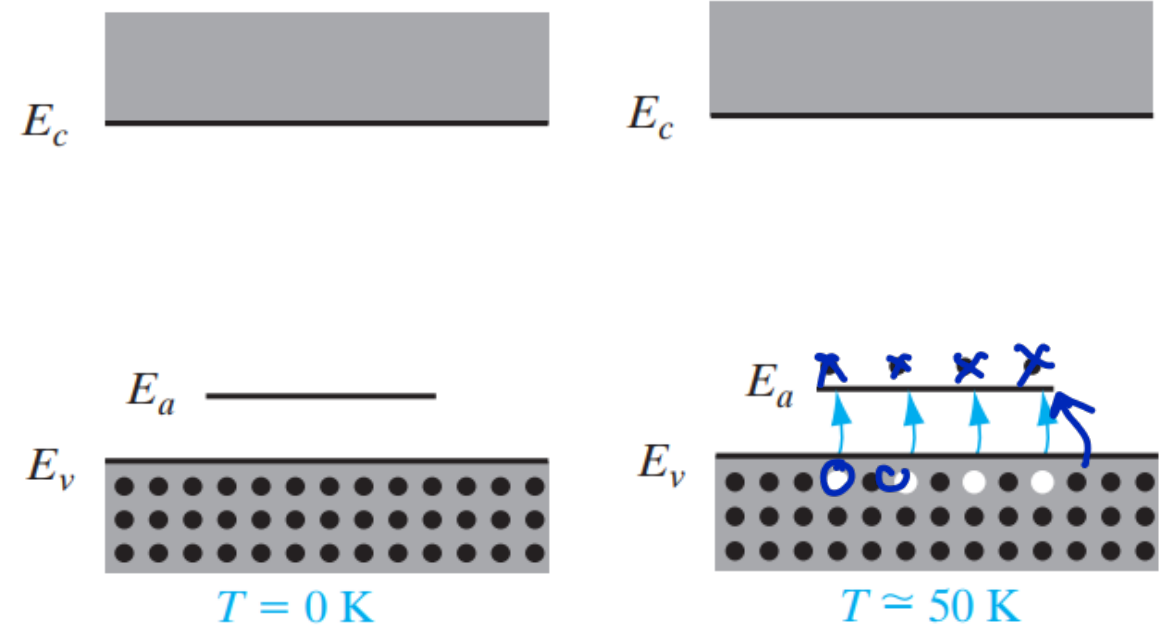
Extrinsic Semiconductor (p-type)



Majority carriers: Holes
Minority carriers: Electrons

Acceptors
Boron (P)
Gallium (Ga)
Indium (In)
Aluminum (Al)

Impurity Atoms
Trivalent



How to calculate the intrinsic carrier concentration (n_i)?
How to find majority carrier concentration?
How to calculate minority carrier concentration?

Doping – How much?

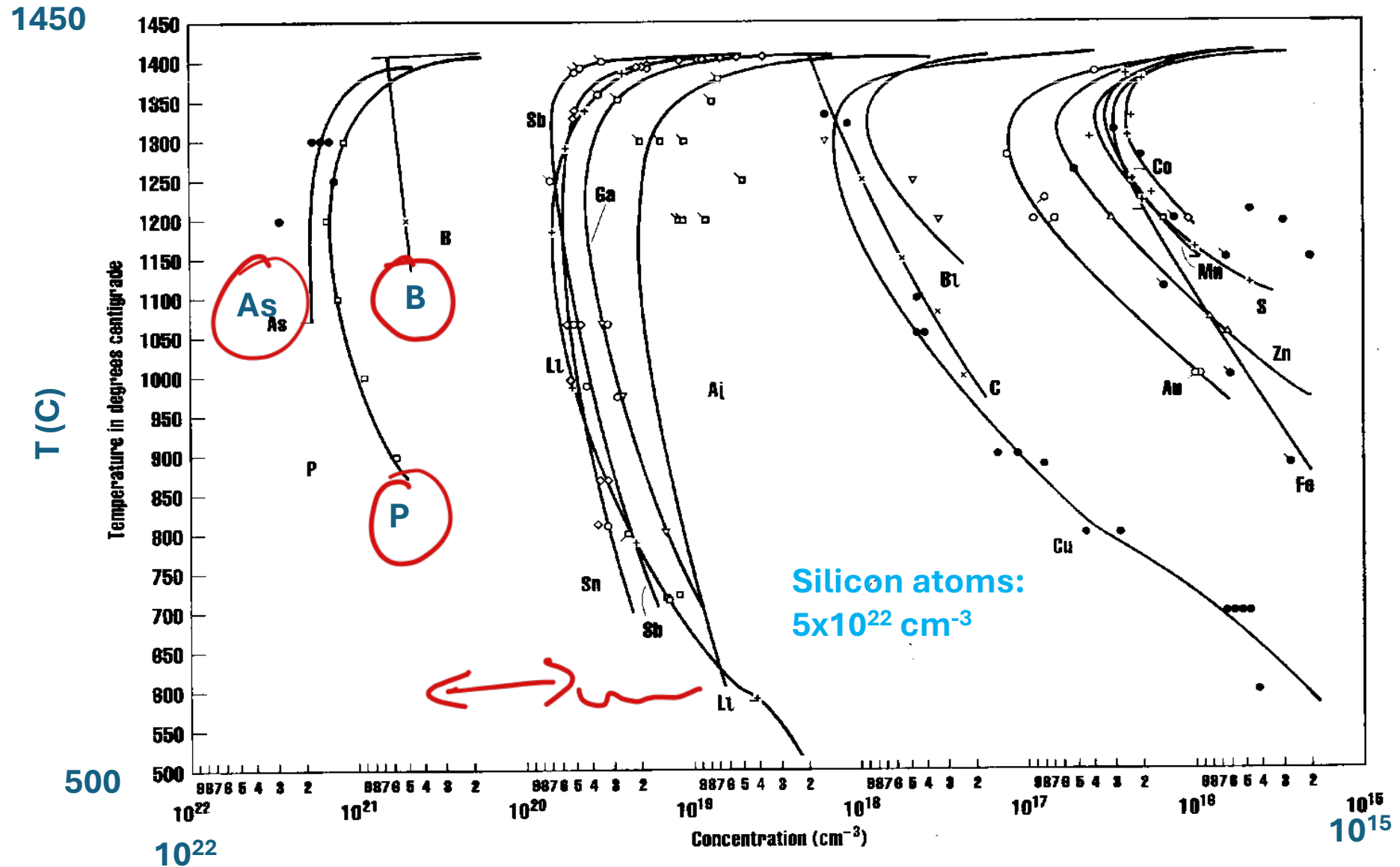


Figure 2-4 Solid solubility of common silicon impurities. (All rights reserved, reprinted with permission, © 1960 AT&T.)

Questions unanswered? Intrinsic & Extrinsic Semiconductor

1. How much is n_i ? (free)
2. How are n_i and Bandgap Related?
3. How many are electrons and how many are holes in the extrinsic semiconductors?
4. Is there any relation between electrons and holes?

Is Electron particle or wave?

Quantum/Wave Mechanics before we move ahead

Schrödinger equation

Wavefunction (ψ)

$$\left[-\frac{\hbar^2}{2m_0} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) + V_0 \right] \psi(r) = E\psi(r)$$

Solving:

The energy:

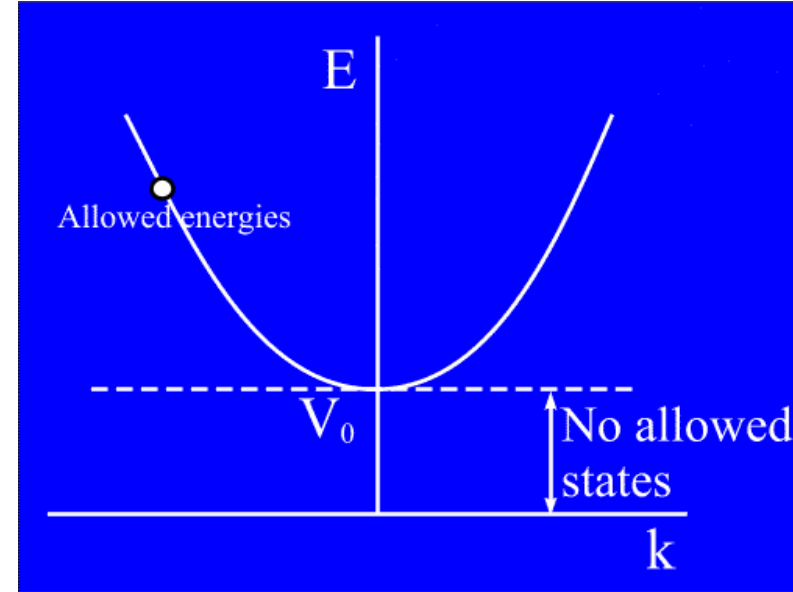
$$E = \frac{\hbar^2 k^2}{2m_0} + V_0$$

Momentum:

$$p = \hbar k$$

k is the wavevector: $k = \frac{2\pi}{\lambda}$

Probability Density ($|\psi|^2$)



$$E = \frac{1}{2}mv^2 = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m_0}$$

For free electrons

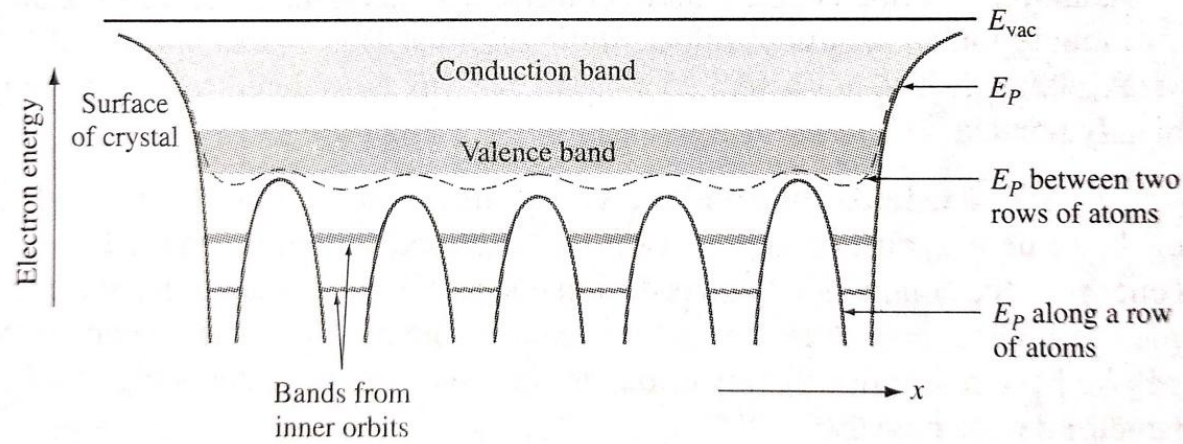
$$\frac{d^2 E}{dk^2} = \frac{\hbar^2}{m^*}$$

m^* = Effective mass of electron
(All QM is added in *)

m = mass of free electron

Finding Charge Carrier Concentration

To understand electrical properties, device behavior it is necessary to find the charge carriers



Need to find distribution of carriers over the available energy states!

1. Density of states (DOS) ✓
2. Probability of finding charge carrier in these states $f(E)$ ✓

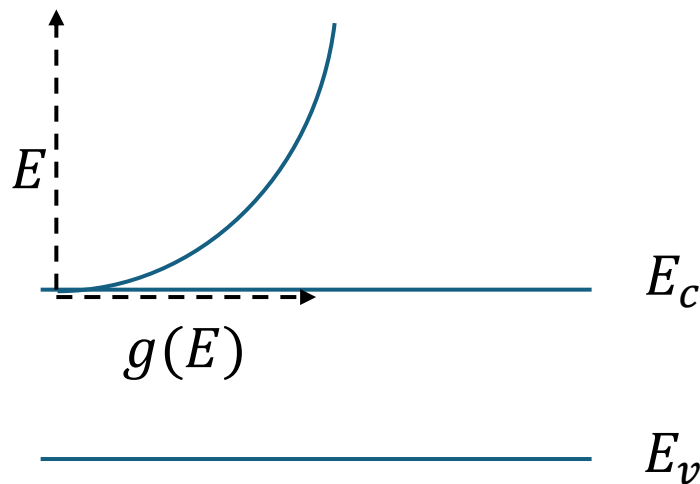
The Density of States (DOS)

Number of states per unit volume per unit energy

$$g(E) \propto \sqrt{E}$$

For electron near conduction band

$$g(E) = \frac{1}{2\pi^2} \left(\frac{2m_e^*}{\hbar^2} \right)^{\frac{3}{2}} \sqrt{E - E_c}$$



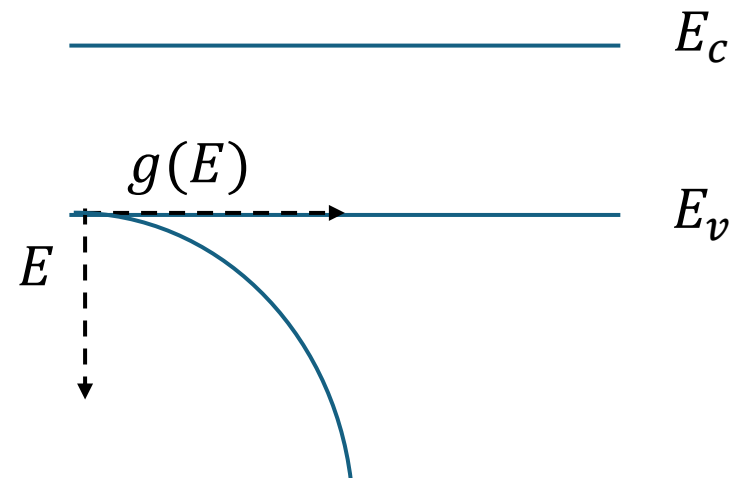
DOS distribution function

For free electron

$$g(E) = \frac{1}{2\pi^2} \left(\frac{2m_o}{\hbar^2} \right)^{\frac{3}{2}} \sqrt{E - E_0}$$

For holes near the valence band

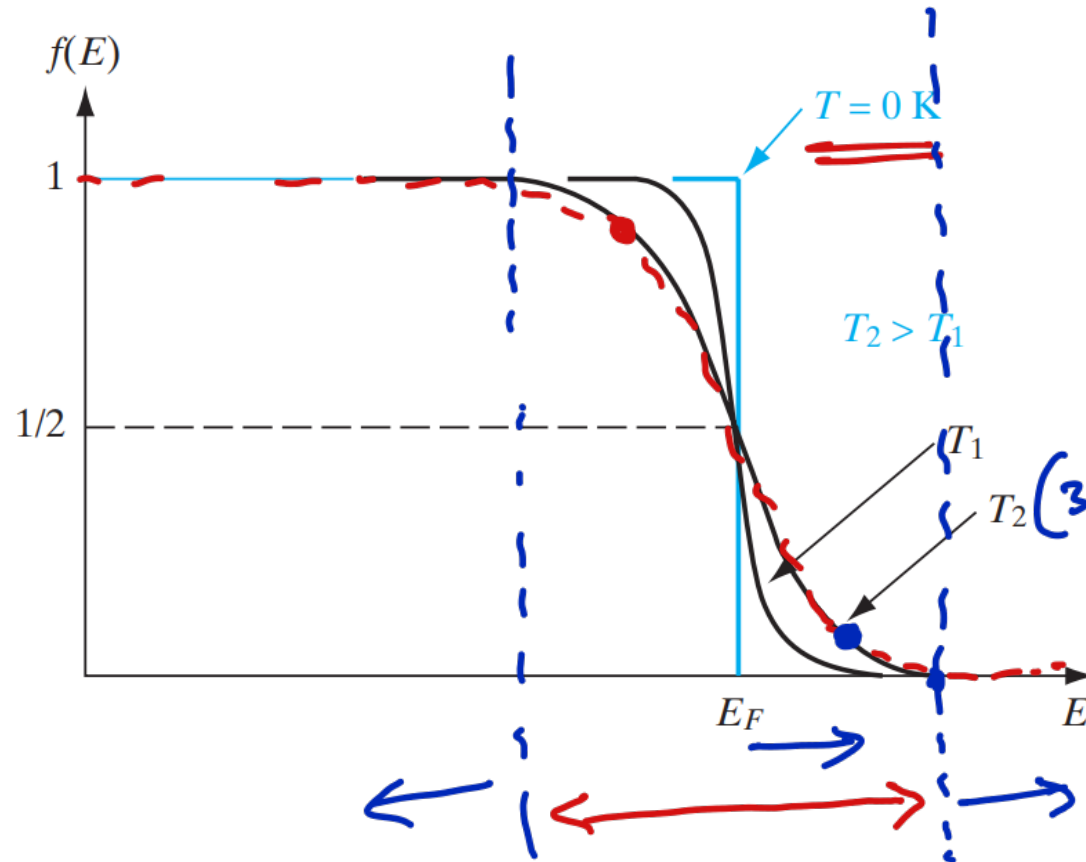
$$g(E) = \frac{1}{2\pi^2} \left(\frac{2m_h^*}{\hbar^2} \right)^{\frac{3}{2}} \sqrt{E_v - E}$$



The Fermi Level

Fermi–Dirac distribution function

⇒ $f(E) = \frac{1}{1 + e^{(E-E_F)/kT}}$



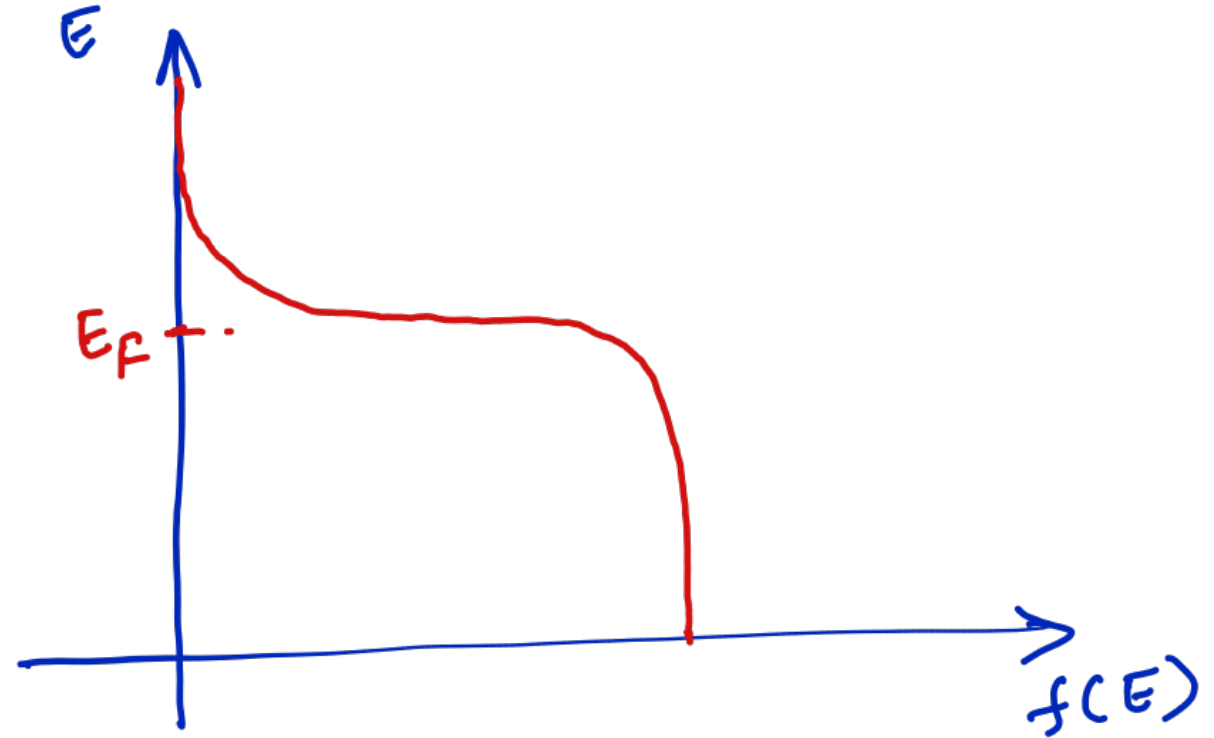
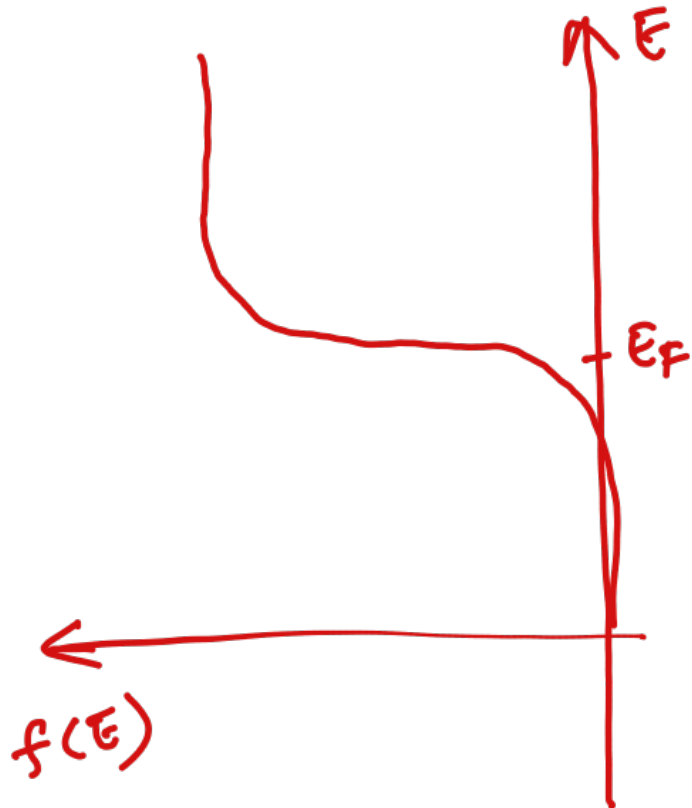
- E_F is known as the fermi Level
- $f(E)$ is the probability of occupancy of an available state at E
- At 0K, every available energy state up to E_F is filled with electrons, and all states above E_F are empty
- Above 0K:
 - Some probability of states being filled above E_F
 - Some probability of states being empty below E_F

$\sim 10^{10} / \text{cm}^3$

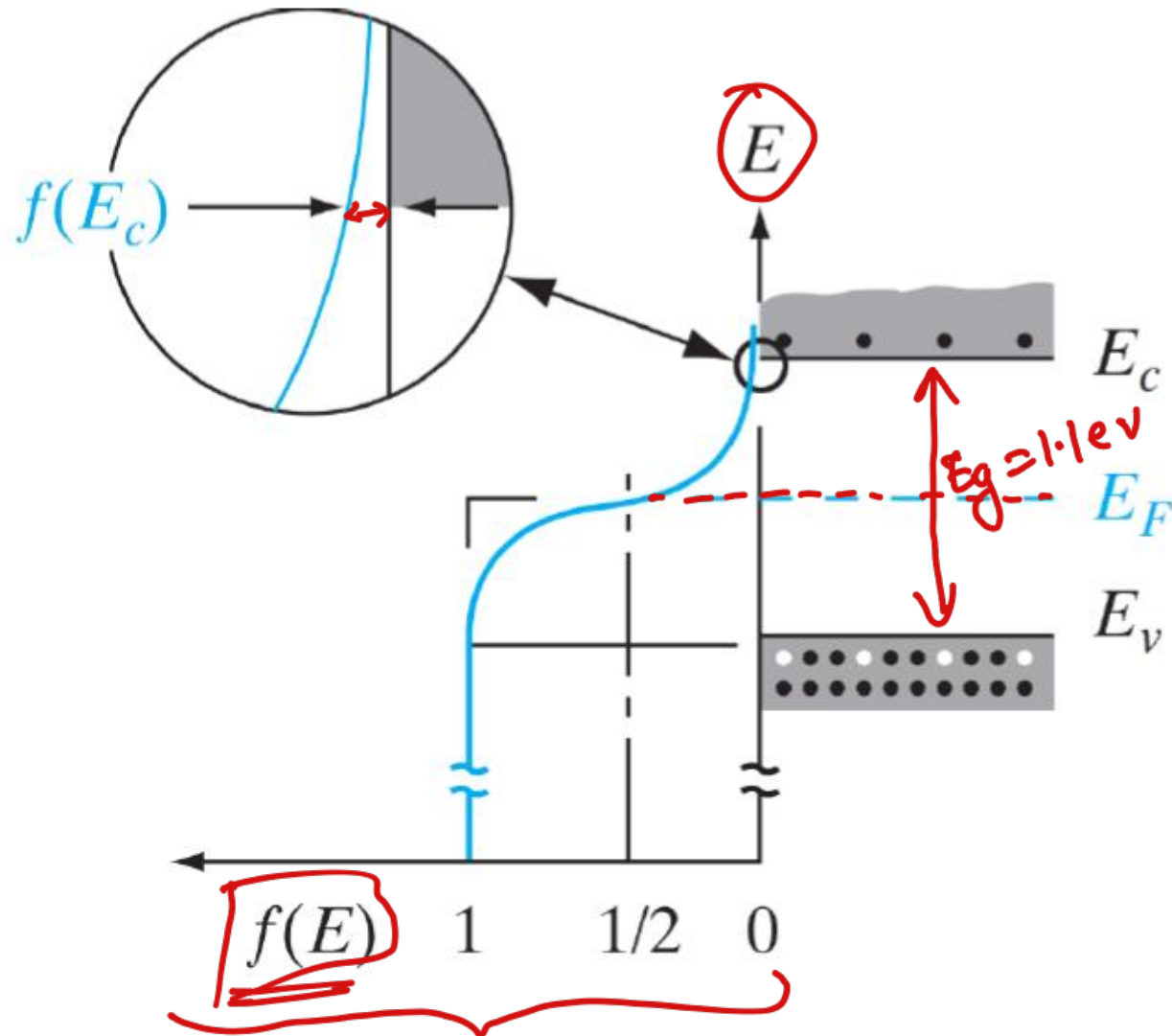
Occupancy Probability – Different ways to plot

Fermi–Dirac distribution function

$$f(E) = \frac{1}{1 + e^{(E-E_F)/kT}}$$



Occupancy Probability – Intrinsic Semiconductor



Recall: $T=300\text{K}$, $V_{th} = 26\text{mV}$, $E_g = 1.1\text{eV}$

How do we have n_i ?

$$V_{th} = 26\text{meV}$$
$$\approx 10^{10} / \text{cm}^3$$

" $1-f(E)$ "

Finding Carrier Concentration – Intrinsic Semiconductor

Density of states

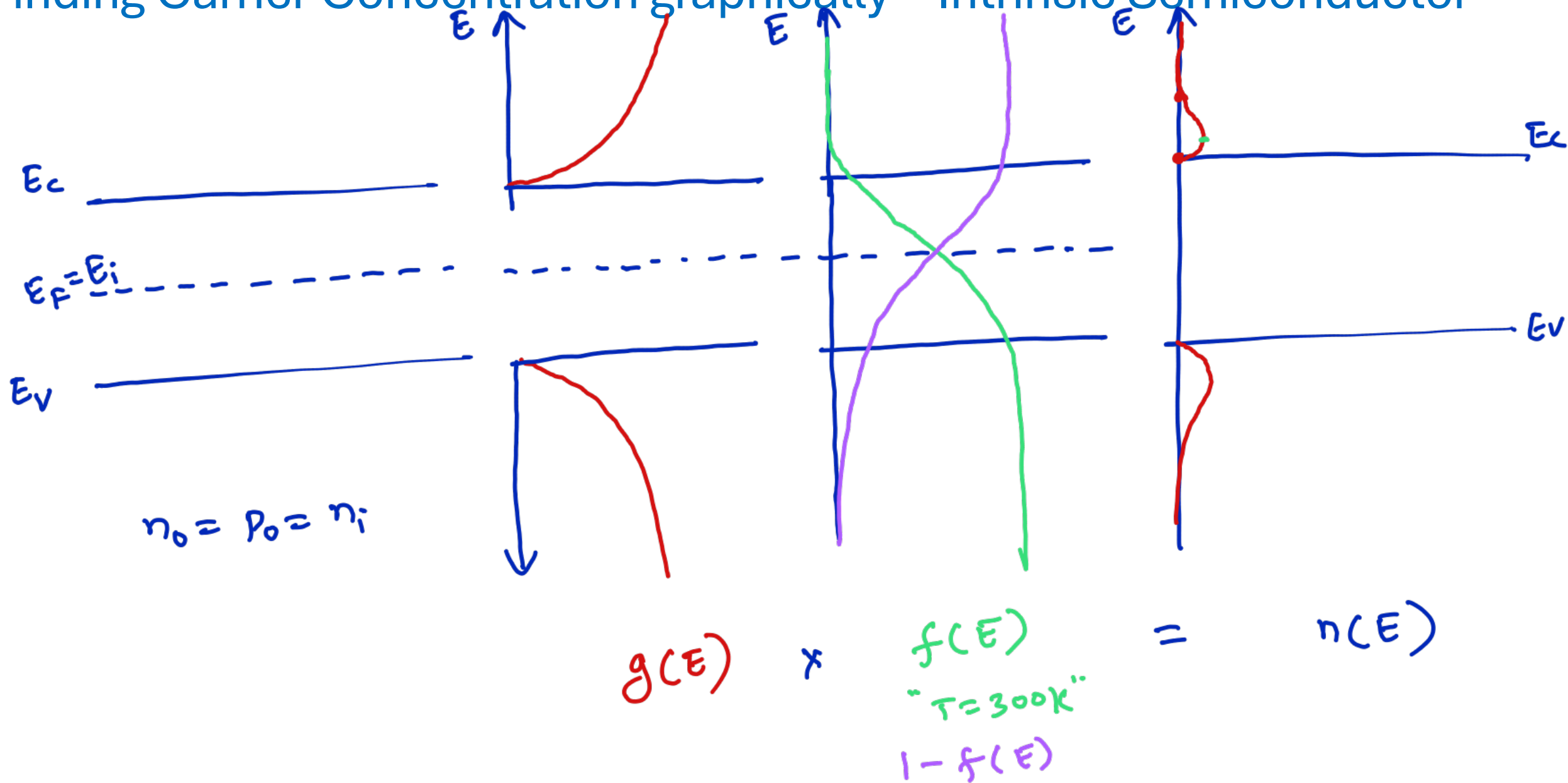
$$\text{For holes: } g(E) = \frac{1}{2\pi^2} \left(\frac{2m_e^*}{\hbar^2} \right)^{\frac{3}{2}} \sqrt{E_v - E}$$

$$\text{For electrons: } g(E) = \frac{1}{2\pi^2} \left(\frac{2m_e^*}{\hbar^2} \right)^{\frac{3}{2}} \sqrt{E - E_c}$$

Fermi–Dirac probability distribution function

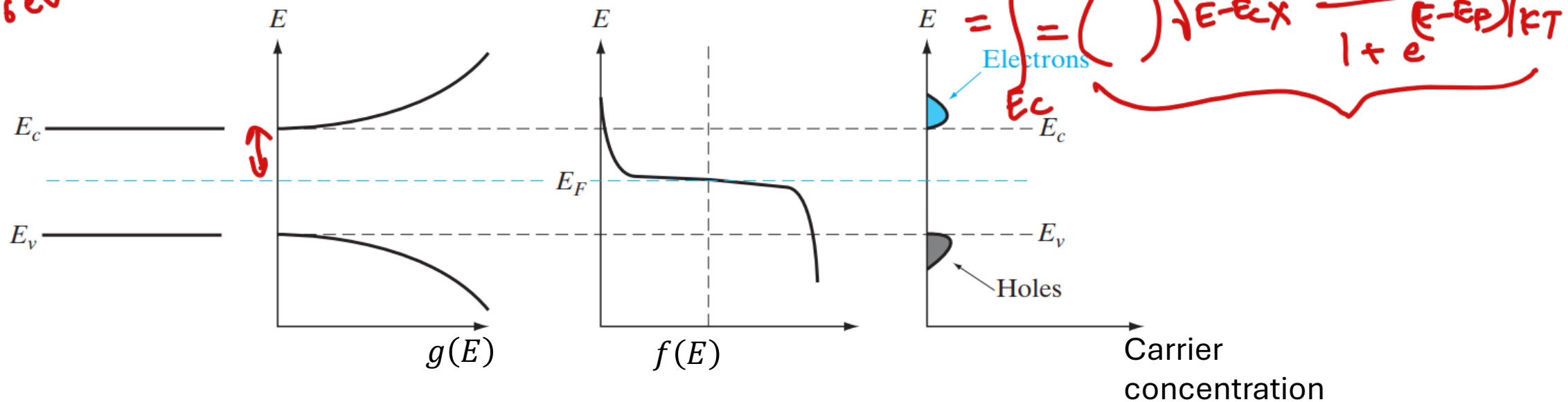
$$f(E) = \frac{1}{1 + e^{(E - E_F)/kT}}$$

Finding Carrier Concentration graphically – Intrinsic Semiconductor



Finding Carrier Concentration – Intrinsic Semiconductor

Si: 1.1 eV
Ge: 0.66 eV



Total electron Concentration in CB ($\#/cm^3$)

$$\boxed{n_0} = \int_{E_c}^{\infty} g(E) f(E) dE = N_C e^{-\left[\frac{(E_c - E_F)}{kT}\right]} \sim 10^{10} / cm^3$$

where $N_C = 2 \left(\frac{m_e^* kT}{2\pi \hbar^2} \right)^{\frac{3}{2}}$ Effective DOS on CB

Total holes Concentration in VB ($\#/cm^3$)

$$\underline{p_0} = \int_{-\infty}^{E_v} g(E) f(E) dE = N_V e^{-\left[\frac{(E_F - E_v)}{kT}\right]}$$

where $N_V = 2 \left(\frac{m_h^* kT}{2\pi \hbar^2} \right)^{\frac{3}{2}}$ Effective DOS on VB

Intrinsic Carrier Concentration – Is it related to bandgap?

$$n_0 = N_c e^{-[E_c - E_F]/kT}$$

$$p_0 = N_v e^{-[E_F - E_v]/kT}$$

$$\boxed{n_0 = p_0 = n_i}$$

$$n_0 p_0 = n_i^2 = N_c N_v e^{-[E_c - E_v]/kT}$$

$$n_i^2 = N_c N_v e^{-E_g/kT}$$

$$\boxed{n_i^2 = N_c N_v e^{-E_g/kT}}$$

$$E_g \uparrow \Rightarrow n_i \downarrow$$

Intrinsic Carrier Concentration

Mass Action Law

$$n_0 p_0 = N_C N_V e^{-\left(\frac{E_g}{kT}\right)} = n_i^2$$

$$N_C = 2 \left(\frac{m_e^* kT}{2\pi \hbar^2} \right)^{\frac{3}{2}}$$

$$N_V = 2 \left(\frac{m_h^* kT}{2\pi \hbar^2} \right)^{\frac{3}{2}}$$

	Si	Ge	GaAs
Energy Gap at 300K (eV)	1.12	0.66	1.424
Intrinsic Carrier Concentration (cm ⁻³)	1.45 x 10 ¹⁰	2.4 x 10 ¹³	1.79 x 10 ⁶

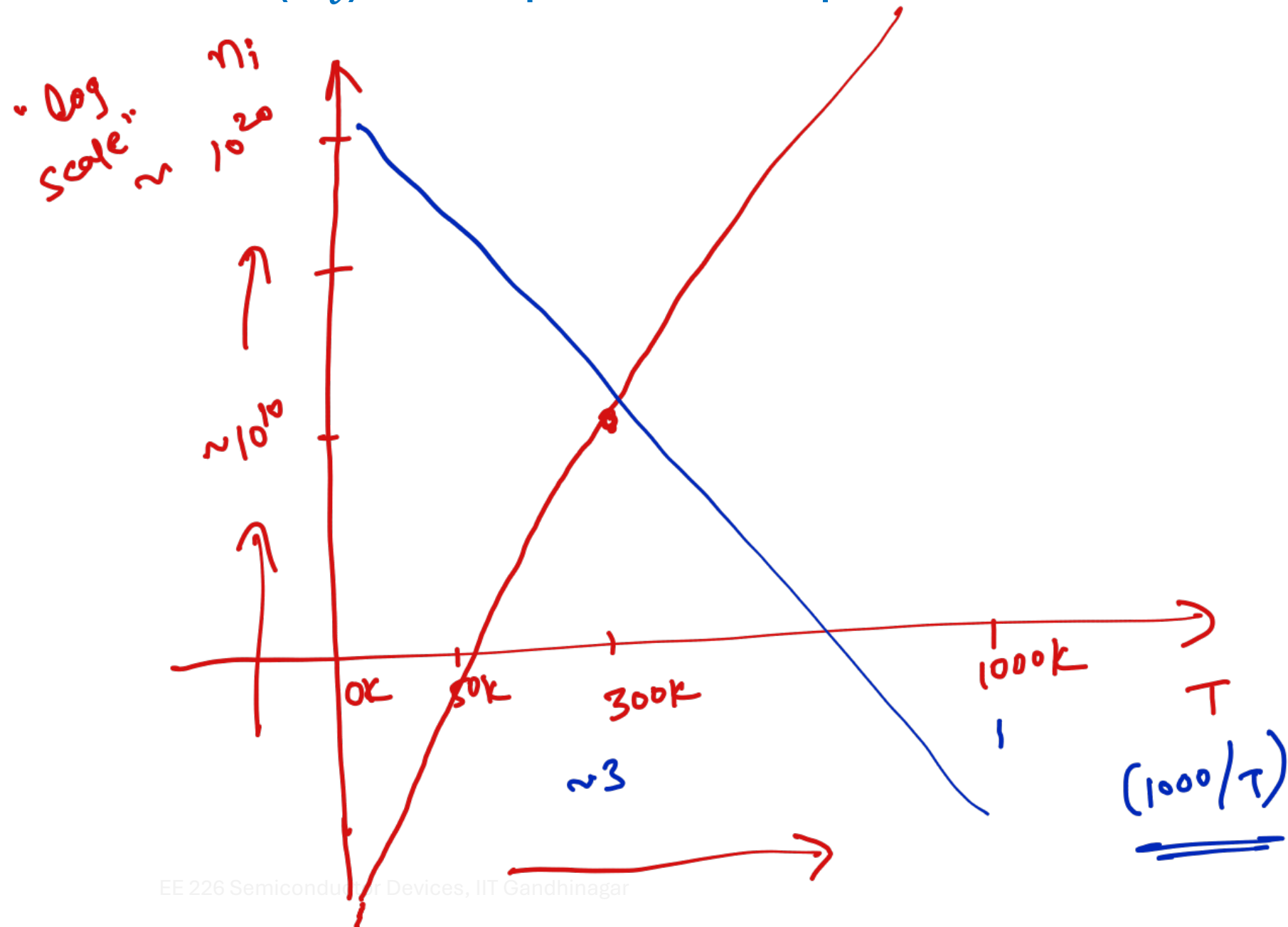
	Germanium	Silicon	Gallium Arsenide
N _c (cm ⁻³)	1.02 x 10 ¹⁹	2.81 x 10 ¹⁹	4.35 x 10 ¹⁷
N _v (cm ⁻³)	5.64 x 10 ¹⁸	1.83 x 10 ¹⁹	7.57 x 10 ¹⁸

Intrinsic Carrier Concentration (n_i) – Temperature Dependence

$$n_0 p_0 = N_C N_V e^{-\left(\frac{E_g}{kT}\right)} = n_i^2$$

$$N_C = 2 \left(\frac{m_e^* kT}{2\pi \hbar^2} \right)^{\frac{3}{2}}$$

$$N_V = 2 \left(\frac{m_h^* kT}{2\pi \hbar^2} \right)^{\frac{3}{2}}$$

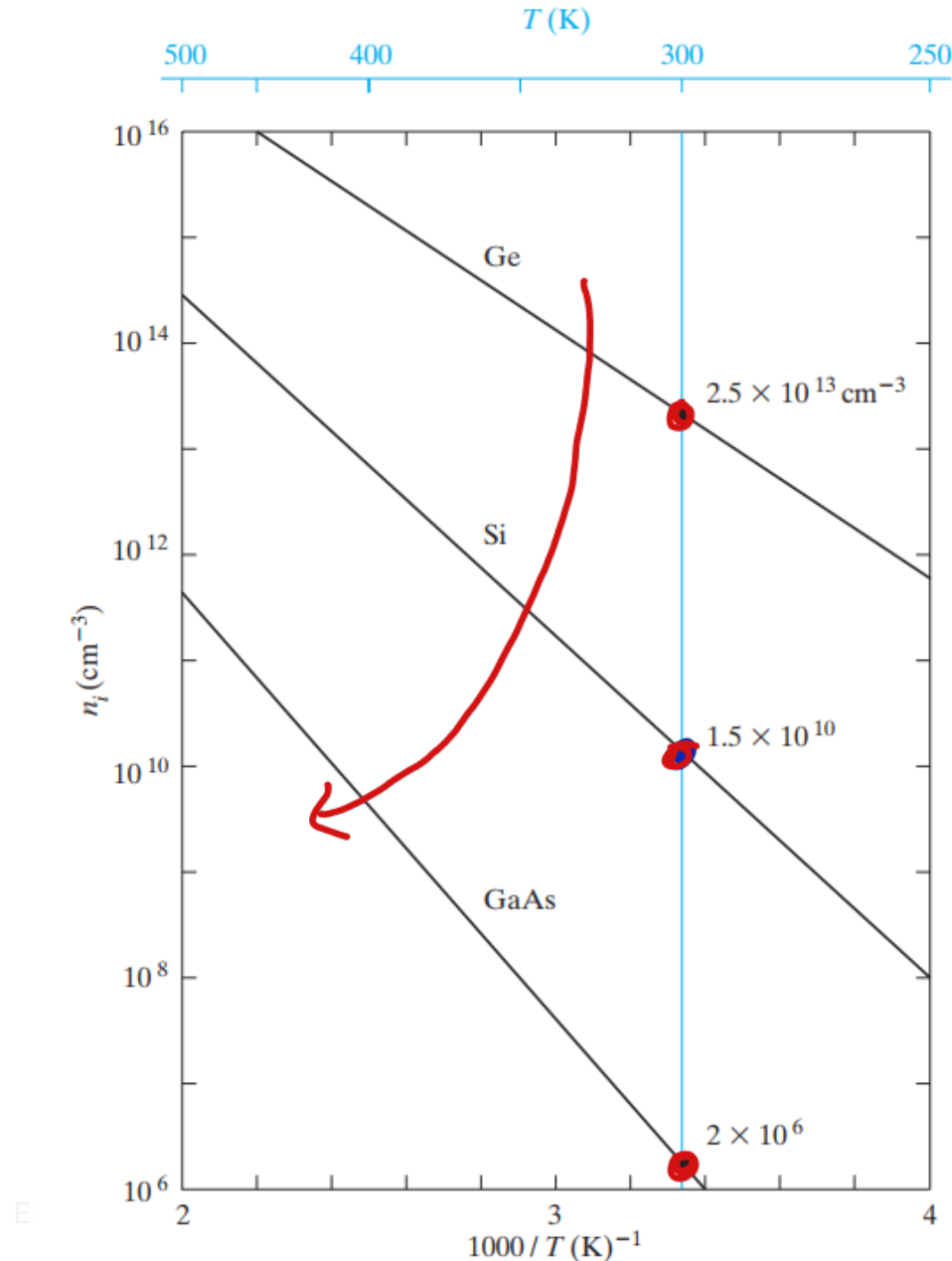


Intrinsic Carrier Concentration (n_i) – Temperature Dependence

$$n_0 p_0 = N_C N_V e^{-\left(\frac{E_g}{kT}\right)} = n_i^2$$

$$N_C = 2 \left(\frac{m_e^* kT}{2\pi \hbar^2} \right)^{\frac{3}{2}}$$

$$N_V = 2 \left(\frac{m_h^* kT}{2\pi \hbar^2} \right)^{\frac{3}{2}}$$



Alternate Expression for intrinsic Carrier Concentration

Electrons

$$n_0 = N_C e^{-\left[\frac{(E_C - E_F)}{kT}\right]} \quad \text{--- (1)}$$

Holes

$$p_0 = N_V e^{-\left[\frac{(E_F - E_V)}{kT}\right]}$$

$$\boxed{n_0 = p_0 = n_i \quad E_F = E_i}$$

$$\boxed{p_0 = n_i e^{(E_i - E_F)/kT}}$$

$$n_i = N_C e^{-(E_C - E_i)/kT}$$

$$\Rightarrow N_C = n_i e^{(E_C - E_i)/kT} \quad \text{--- (2)}$$

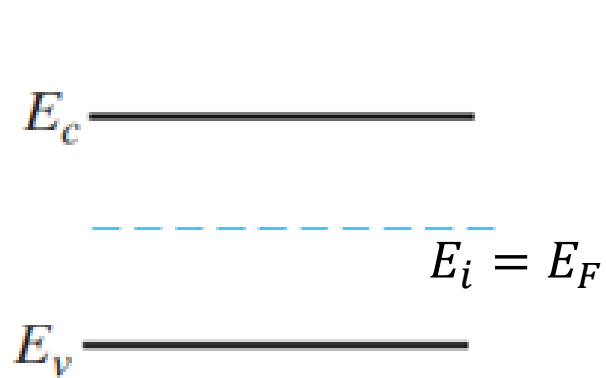
Put (2) in (1)

$$\Rightarrow n_0 = n_i e^{(-\cancel{E_C} + E_F + \cancel{E_C} - E_i)/kT}$$

$$\boxed{n_0 = n_i e^{(E_F - E_i)/kT}}$$

$$n_0 = n_i e^{(E_i - E_F)/kT}$$

Intrinsic Semiconductor – Where is E_i (i.e. E_F)?



$$n_0 = p_0 = n_i \Rightarrow N_C e^{-\left[\frac{(E_C - E_i)}{kT}\right]} = N_V e^{-\left[\frac{(E_i - E_v)}{kT}\right]}$$

$$\frac{N_V}{N_C} = \left(-E_C + E_i + E_i - E_V \right) / kT$$

$$\ln\left(\frac{N_V}{N_C}\right) = \frac{2E_i}{kT} - \frac{E_C + E_V}{kT}$$

" E_i will be slightly above center"

$$N_C = 2 \left(\frac{m_e^* kT}{2\pi\hbar^2} \right)^{\frac{3}{2}}$$

$$N_V = 2 \left(\frac{m_h^* kT}{2\pi\hbar^2} \right)^{\frac{3}{2}}$$

$$m_h^* > m_e^*$$

$$\Rightarrow E_i = \frac{E_C + E_V}{2} + \frac{kT}{2} \ln\left(\frac{N_V}{N_C}\right)$$

$$E_i = \frac{E_C + E_V}{2} + \frac{3kT}{4} \ln\left(\frac{m_h^*}{m_e^*}\right) \leftarrow \text{"mev"}$$

Alternate Expression for intrinsic Carrier Concentration

Electrons

$$n_0 = N_C e^{-\left[\frac{(E_C - E_F)}{kT}\right]}$$

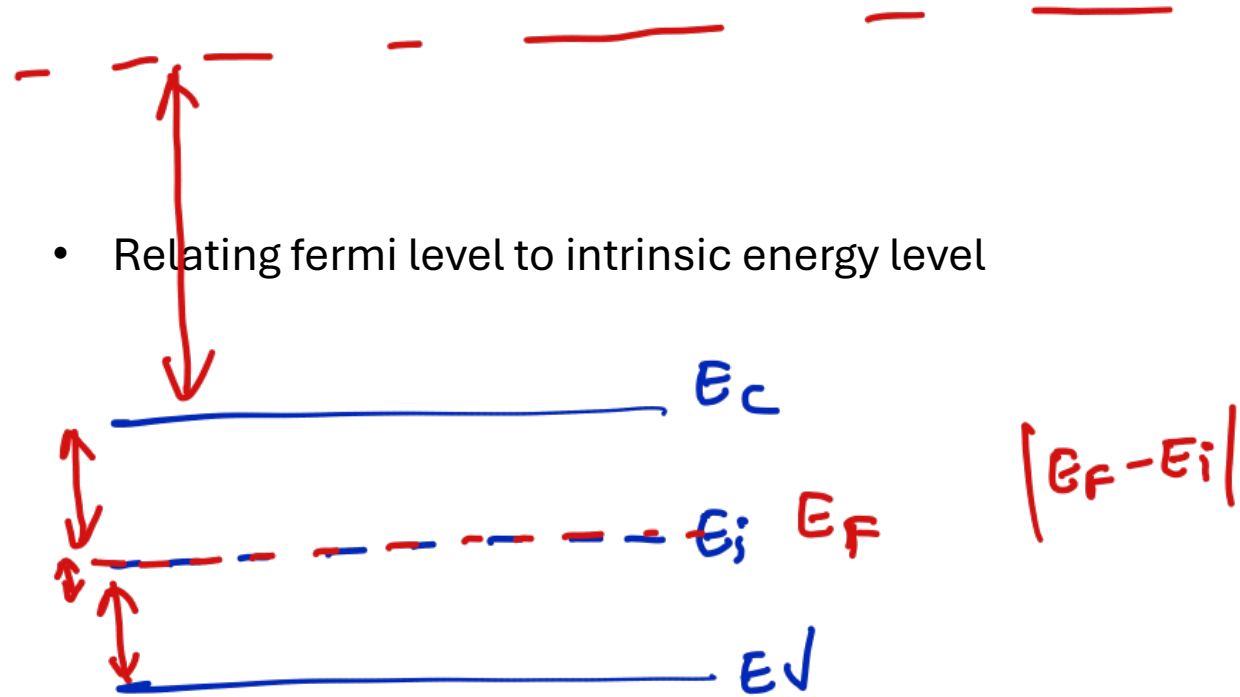
$$\Rightarrow n_0 = n_i e^{\left(\frac{E_F - E_i}{kT}\right)}$$

$$n_0 = n_i = N_C e^{-\left[\frac{(E_C - E_i)}{kT}\right]}$$

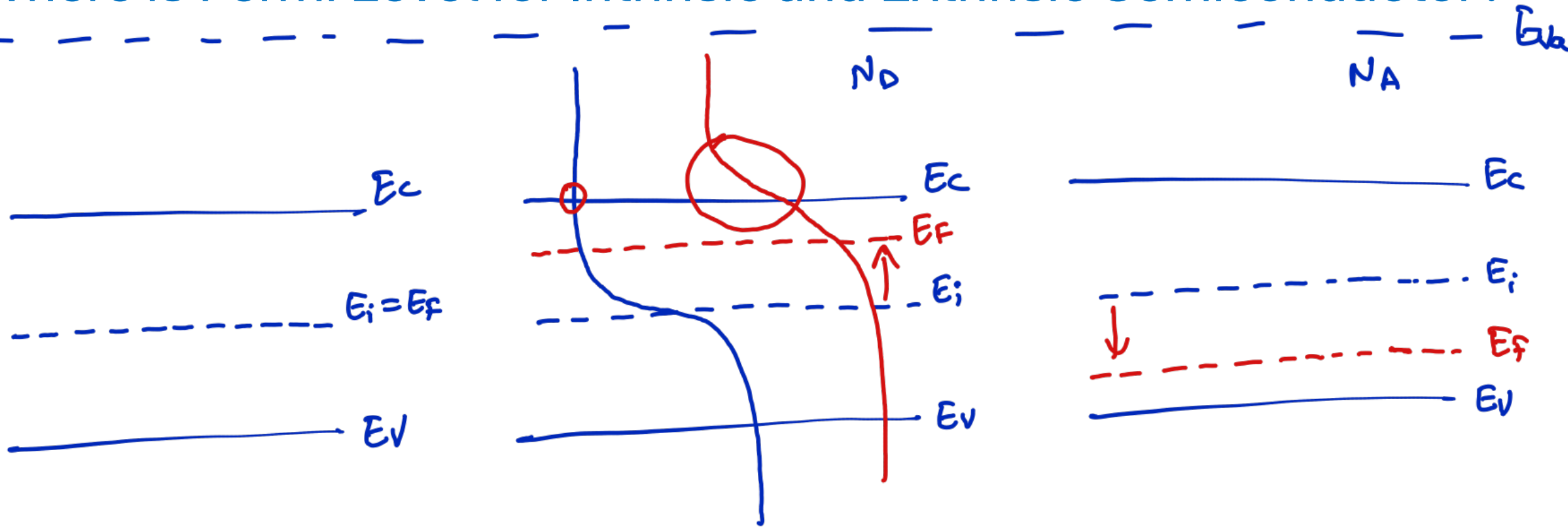
$$\text{Rewriting: } N_C = n_i e^{\left[\frac{E_C - E_i}{kT}\right]}$$

Holes

$$p_0 = n_i e^{\left[\frac{(E_i - E_F)}{kT}\right]}$$



Where is Fermi Level for Intrinsic and Extrinsic Semiconductor?



intrinsic

n-type

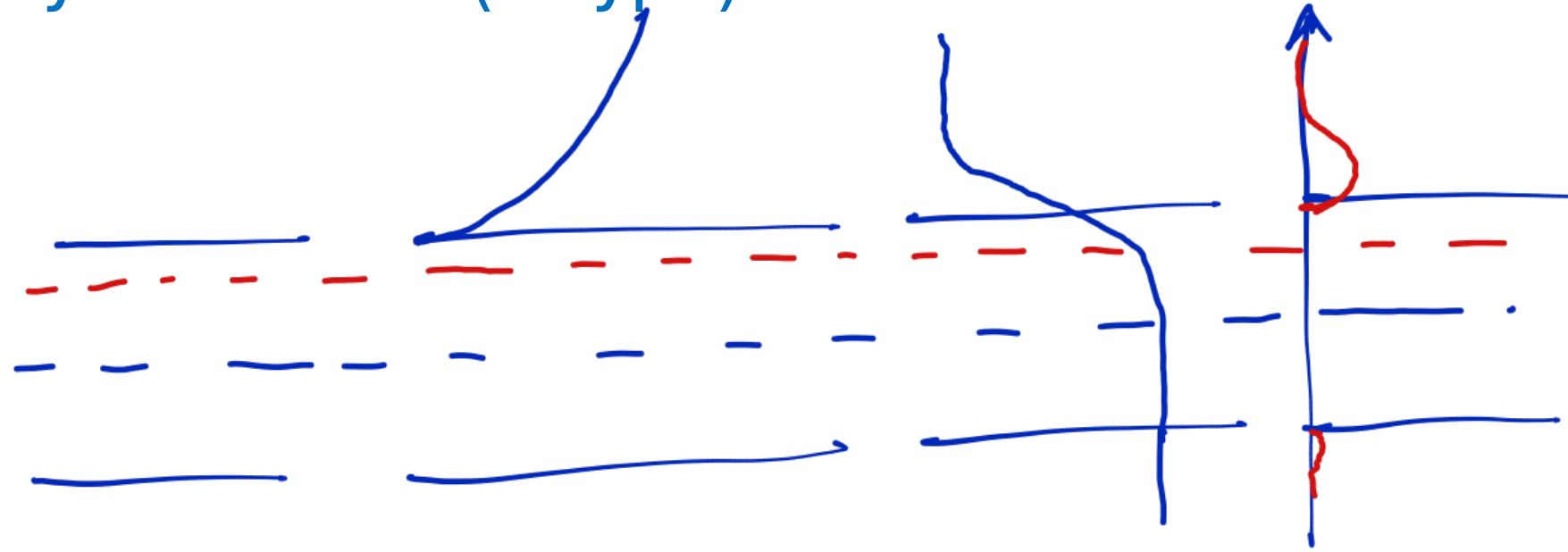
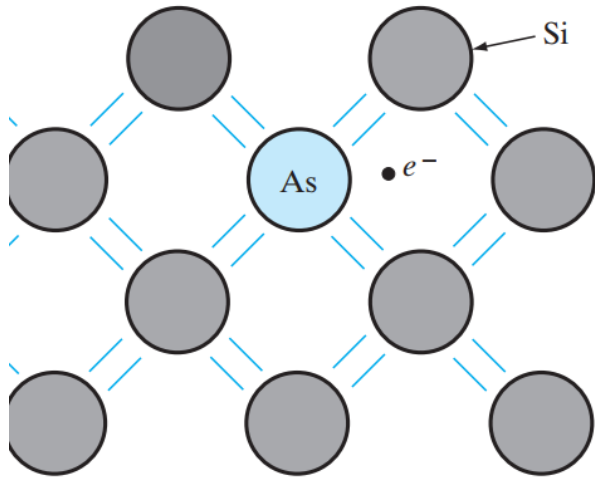
p-type

$$\Rightarrow n = N_D + n_{oh}$$

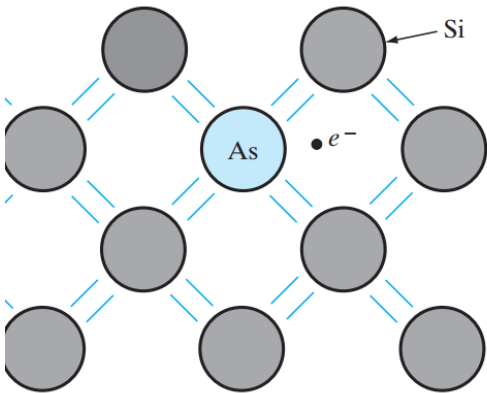
$$p = p_o \downarrow \Rightarrow 10^{10} \rightarrow 10^2$$

$10^8 + 10^{10}$
 10^{10}

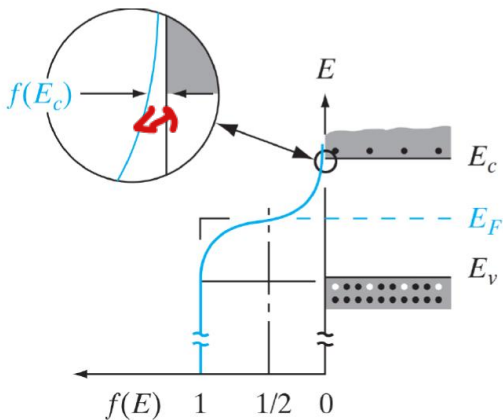
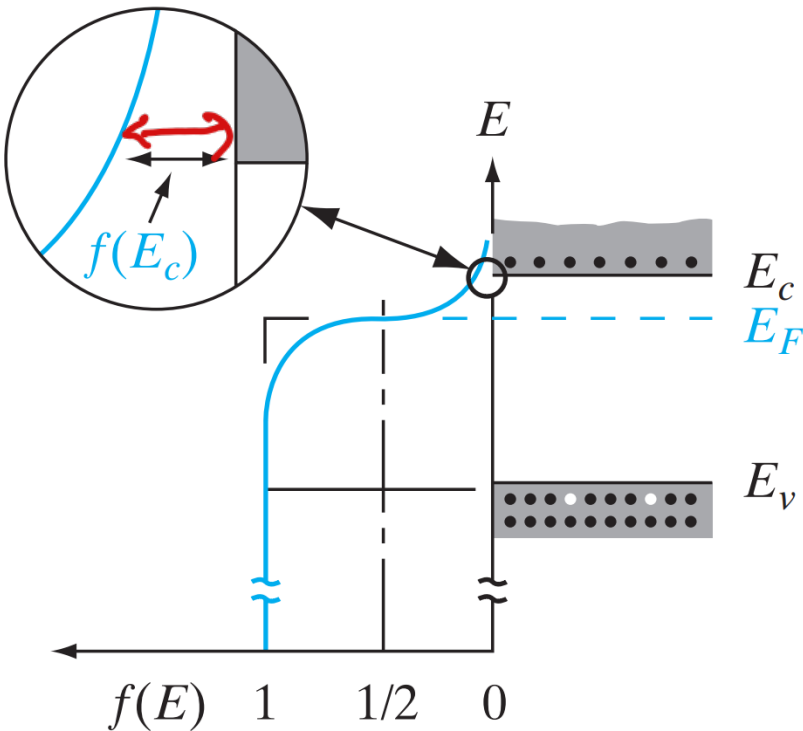
Occupancy Probability – Extrinsic (n-type) Semiconductor



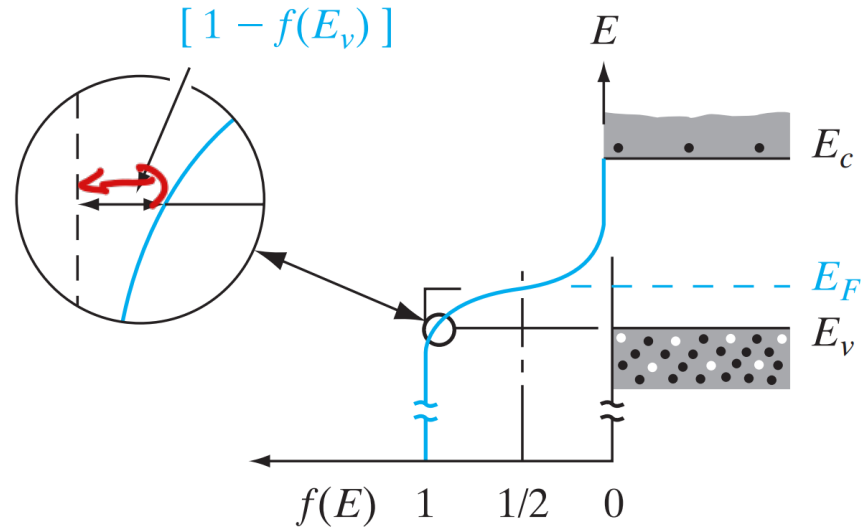
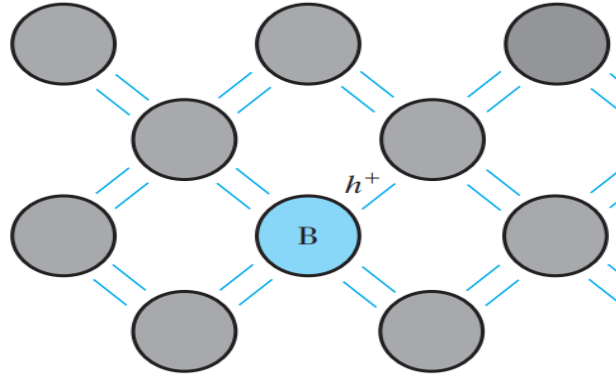
Occupancy Probability – Extrinsic (n-type) Semiconductor



Intrinsic Semiconductor



Occupancy Probability – Extrinsic (p-type) Semiconductor



Finding Carrier Concentration – Extrinsic Semiconductor

$$n_0 = \int g(E) f(E) dE$$

